University of Padua

Bachelor's Degree in Computer Engineering

Lecture notes

Computer Engineering Fundamentals

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Preface

These notes follow the course syllabus and complement the lectures with worked examples and self-assessment exercises.

Revision history

Version	Date	Changes
0.1.0	10 July 2026	Created the initial course outline
0.2.0	12 July 2026	Added the processor organization chapter
0.3.0	15 July 2026	Expanded the instruction-cycle examples
0.4.0	18 July 2026	Added the first memory hierarchy diagrams
0.5.0	20 July 2026	Revised the cache mapping explanation
0.6.0	22 July 2026	Added exercises on address translation
0.7.0	24 July 2026	Expanded the input and output chapter

Version	Date	Changes
0.8.0	26 July 2026	Added the interrupt-handling examples
0.9.0	28 July 2026	Reviewed terminology and cross-references
1.0.0	1 August 2026	Published the introductory lectures
1.1.0	2 August 2026	Added the memory hierarchy summary
1.2.0	3 August 2026	Corrected examples and improved navigation

1 Introduction to computer systems

Figure 1.1. Functional organization of a computer

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A Exam reference sheet

This appendix summarizes binary prefixes, base conversions, and characteristic memory hierarchy timings.